Appendix A: Crime threats identified through the scoping review and sandpit event (highlighted)

#	*Crime threat*	Description
1	Non-carbon credit frauds	Fraudulent selling of materials and products (e.g. sale of adulterated critical minerals for CCMT development or misrepresentation of the quality/origin of energy products and sources); fraudulent certification schemes for the extraction of natural resource; investment frauds based on bogus CCMT projects.
2	Carbon credit frauds	Making and selling of fraudulent carbon offset certificates; selling carbon credits for projects that do not exist; false verification or overstatement of carbon offsets; fake registries; double selling of carbon credits. These scenarios can be facilitated by laissez-faire environments and poor government controls in carbon/green credit markets.
3	Money laundering	Money laundering by buying carbon credits with "dirty" money.
4	Carbon and energy market abuse and manipulation	Unfair distribution/allocation of carbon emission allowances in the carbon trading market can lead to power abuse. The carbon market is also susceptible to manipulation due to inconsistent/lack of standardised measurements of carbon emissions and offsets. There also may be unfair competition due to monopolistic tendencies of electricity distributors/suppliers that may disallow smaller local green energy suppliers from becoming operational.
5	Greenwashing or disinformation	Any intentional practices by organisations to make them appear more sustainable or environmentally friendly than they truly are to gain profits or other benefits (e.g. reputational advantage). For example, corporations investing in the voluntary carbon market may convey a false appearance of ecological welfare to the public and stakeholders, using CCUS to distract from climate change impact of fossil fuel reliance and emissions from production processes, misrepresenting the nature of the energy used, communicating misleading estimates of greenhouse gas emissions and reductions, and spreading disinformation relating to CCMTs. (Not all these practices are criminal per se. The criminal nature of these practices depends on their relationships with other behaviours – for instance, deliberate misrepresentation to deceive investors or buyers about the quality of a product or service can amount to fraud.)
6	Corrupt practices	Many corrupt practices can cause inappropriate allocation and wastage of resources (time, money, materials) and compromise sustainable development projects and the quality of renewable energy infrastructure. Examples include: companies/investors bribing public officials to unduly acquire licenses

		or procurement contracts for CCMT projects (e.g. energy plants) also to the detriment of other bidders; companies/investors bribing political leaders to support dependency on fossil fuels or halt sustainable developments; governments or companies misappropriating international funds awarded to develop CCMTs; rogue companies getting subsidies for sustainable development projects without making actual progress or delivering poor quality work (rent-seeking); public officials turning a blind eye on carbon leakages.
7	Stalking and harassment	Stalking ex-partner through internet-of-things devices required for smart/sustainable energy management; anti-climate change individuals/groups harassing developers or installers of CCMTs.
8	Handling of stolen goods	Selling or otherwise illicitly disposing of stolen materials or technologies (e.g. copper and cables).
9	Sabotage and criminal damage	Hostile states or organised criminal groups may disrupt energy supply, balancing and distribution by attacking physical energy infrastructure nodes, including by operational technology, or attacking the transportation infrastructures of raw materials or technology components.
10	Illegal occupation and control of critical resources	Hostile and violent state or non-state actors occupy or otherwise control critical resources (e.g. mining sites) for CCMT development, preventing access and utilisation by legitimate actors.
11	Blackmail or extortion	Countries storing carbon may threaten to release all of the carbon from their reservoirs unless their "demands were met" by another country. A state may also (threaten to) stop energy supply to another state, for example, in response to war, tariffs or sanctions.
12	Vandalism	Physically damaging public clean/renewable energy infrastructure/assets and misuse of "clean cooking" equipment by local community meant to transition to a cleaner lifestyle, particularly by individuals that may be unhappy with new environmental regulations.
13	Tax fraud and tax avoidance	Buying carbon credits in low-tax countries and selling them in high-tax countries without paying the tax difference to the state; inflating innovation input costs in order to obtain tax relief; abuse of the Carbon Capture Usage and Storage (CCUS) tax credit system abused to avoid tax; creation of fraudulent companies providing fake CCMT certificates to exploit existing government incentives and tax credits for the installation of CCMTs in private homes.
14	Theft of technology, materials or energy	Theft of various CCMTs, their components or related resources from construction sites, plants and installations (e.g. electric vehicles, batteries, second-hand lithium, copper, cables, solar panels and vehicles near solar farms to transport stolen goods); energy theft by changing current on charging point, disconnecting the voltage link or connecting the voltage divider resistor in series on the voltage coil, short-

		circuiting the incoming and outgoing wires of the live wire inside the watt-hour meter through copper wires.
15	Public order offences	Anti-climate change or anti-CCMT sentiments may motivate public disorder offences such as affray, riots, violent protests and the incitement of such offences (online or offline). Operators may equally respond violently to such actions. Hostile states and actors may launch influence campaigns to incite civil unrest and sowing disunity related to climate change.
16	Territorial conflict	Offshore wind farms and ocean energy installations may provoke conflicts over territory/use of space, e.g. in case of Taiwan.
17	Terrorism	Nuclear weapon proliferation (i.e., more available nuclear (fusion) reactors increase risk of proliferation); using radioactive material in a suicide attack and physically attacking carbon reservoirs or critical infrastructure with the intent to wreak havoc (e.g. terrorist organisations that create explosions in areas surrounding CO2 pipelines).
18	Dispossession and displacement	Treating countries in the global South and other peripheries as carbon dump under the guise of needing big new infrastructure projects for sustainable transition, de facto involving land enclosures e.g. for biofuel plantations and extracting as much resources, energy and profits as fast as possible (i.e., "green extractivism"); encroaching land/land grabbing from local and indigenous communities for "green energy projects" (e.g. cases reported on Israeli land grabbing from Palestinians); international investors trying to establish mining site by displacing people and dispossessing land (e.g. case in north Portugal).
19	Human trafficking	Deporting people to force them to work in factories and mining sites required for CCMT development.
20	Slavery, forced and illegal labour	Forced or illegal labour in mining sites for the extraction of resources required by CCMTs; mistreatment of female workers and forced labour in solar panel factories.
21	Child labour	Children working in supply chain of electric vehicles and renewable energy technologies.
22	Illegal fishing in blue carbon zones	Blue carbon zones (protected ecosystems meant to provide a safe haven for marine species that naturally include a variety of fish) may attract illegal trawling and fishing.
23	Illicit disposal and handling of waste, energy, material or technologies	Discarding solar panels to landfills; disposal of carbon-based/outdated tech (e.g. refrigerators); illegal waste dumping; failing to ensure proper waste disposal after selling waste to third parties.
24	Computer misuse to breach, intercept or tamper with data or systems	Hacking or misuse of a climate change mitigation technology to breach, tamper with or intercept the data or system of the technology itself – e.g. interfering or stealing private user data from a vehicle at a charging

		point/grid system exploiting system vulnerabilities, or accessing valuable intellectual property stored on a CCMT device.
25	Illegal trade and smuggling of natural resources	Illegal trade and smuggling of precious minerals and other natural resources required for CCMT development. For example, organised crime groups may smuggle raw materials by corrupting port authorities or by exploiting weak surveillance.
26	Illegal extraction of natural resources	Illegal extraction of precious minerals and other natural resources required for CCMT development (e.g. illegal extraction of seabed minerals by companies or state actors).
27	Water insecurity	Carbon capture, utilisation and sequestration may impose new (fresh)water stresses due to additional water requirements for chemical and physical processes to capture and separate CO2. The parasitic loads imposed by carbon capture on power plants will reduce their efficiency and thus require more water for cooling the power plant. Although this scenario might not be necessarily a crime, the scarcity of freshwater it causes may facilitate violent conflict over the resource.
28	Accidents, attacks and harms related to nuclear energy	Risk of accidents or targeted attacks in both the radioactive waste and the reactors, and nuclear waste storage/disposal all may lead to severe environmental damage. As opposed to terrorist scenarios (see "Terrorism"), these may be unintentional, but they may be still criminal in nature (especially if caused by gross negligence or recklessness or by the violation of legal requirements (e.g. environmental or health and safety laws).
29	Environmental harm from mining activity	Extraction processes spoil natural landscapes; cobalt mining (needed for electric devices/vehicles) may lead to harmful air pollution; inappropriate or unlawful exploitation of natural resources may impoverish the environment.
30	Environmental harm from CCUS	Carbon capture, utilisation and sequestration (CCUS) may lead to groundwater contamination due to CO2 leakage during geologic sequestration, acidification of open waters due to CO2 leakage, increased presence of toxic metals due to CO2 leakage and increased N2O emission from CCUS in power plants.
31	Renewable energy projects harm (protected) life	Offshore wind farms in or near marine protected areas can harm ecological life and sea views. Protected bird species have died flying into wind turbines.
32	Forgery/counterfeiting	Falsification of documents/statements to fabricate eligibility onto government subsidy schemes (e.g. falsification of the number of completed works by CCMT installers to unlawfully claim tax relief/credit).
33	Obtaining services dishonestly	Dishonestly using any CCMT-related service or infrastructures without paying for it, e.g. pumping energy into smart grid without paying the required fees.

34	Intellectual property infringements	Research and development into state-of-the-art CCMTs may encourage espionage and IP infringements on said technologies, e.g. to facilitate theft and replication.
35	Computer misuse to enable other offences	Hacking or direct misuse of a climate change mitigation technology to commit other offences. For example: hacking of CCMTs to cause power outages (blackouts) and inconvenience in remote areas with few charging points; tamper with battery system to cause physical/property damage; enable large-scale electric vehicle theft by hacking into a mobile user app; tamper with charging point to tap more energy than paid for and/or launch impersonation attack to charge other users for the energy consumed by the attacker; exploit insecure electronic carbon market registration system to commit cyber theft and financial scams, and prevent people from normally using platform services.

Appendix B: Crime threats identified through the scoping review and sandpit event (highlighted)

#	Countermeasure	Description
1	Economic policies to improve the fairness and integrity of the energy market	Reducing the level of energy market segmentation, breaking the monopoly of ownership and strengthening supervision to reduce the rent-seeking behaviour of local governments and state-owned enterprises. Reducing the government's excessive intervention in energy prices, opening up the energy market and achieving equal competition between private enterprises.
2	Technology and industry regulation	Mandatory/voluntary industry protocols and standards for the cybersecurity of relevant CCMTs, e.g. security key agreement protocol; corporate compliance and Corporate Environmental Responsibility (CER) frameworks; policies defining environmental safety of CCMTS (e.g. offshore wind sector). Clear regulatory definitions of relevant terminology (e.g. what amounts to CCMTs) to prevent misrepresentation; regulation that standardises connections to the electric grid or EV charging points, regulatory requirements to enforce cybersecurity measures for smart meters, regularly updated cybersecurity code of practice including, for instance, mandatory periodical technical controls by sector regulator in consultation with a national technical body, and technical guidance on specific networking and architecture risks for solar and wind production facilities, e.g. accounting for supply chain risks.
3	Policies for the security of and equal access to natural resources	Policies to follow emission pathways with low levels of nuclear and carbon capture, such as CP1-REN, to mitigate the risks of water insecurity. Policy frameworks to ensure that vulnerable and marginalised groups have access to benefits from sustainable energy transition; allowances and auditing of water usage.
4	Environmental law and monitoring/enforcement systems	Giving local forest communities property rights over the trees, better Community Forest Associations (CFA) leadership. Policies/laws to exclude marine protected areas with high biodiversity and sensitive and threatened species and habitats from offshore wind farm development. Regulation of appropriate resource extraction methods to reduce the degradation of the environment and carbon emission.
5	Data collection and sharing for research, policymaking, prevention or investigation	Open-source data to improve detection model development, global coordination of data collection and sharing to identify trends and changing dynamics in relation to criminal threats impacting CCMTs, link datasets across schemes and stakeholders to automatically reject measures that breach

		scheme rules, e.g. subsidies/previous payments on other schemes, utilising satellite imagery and data to track and verify carbon projects and avoid fraud cases.
6	Risk assessment and management	Voluntary or mandatory industry processes and practices to identify, assess and effectively mitigate crime risks related to the development, commercialisation and operation of CCMTs. Conducting corruption risk assessments in the development processes and use of CCMTs.
7	Physical security measures	Surveillance cameras, wheel locks, battery theft detection models, monitoring devices connected to alarm response centres, optic fibre detection systems, systems and devices to detect and intercept anomalous activities in maritime environments.
8	Awareness-raising, public engagement and information	Promoting awareness about crime threats in the renewable energy sector to encourage target hardening. Companies selling CCMTs could highlight crime risks to clients at the point of purchase. Electric car and charger manufacturers, and energy providers should work with industry associations to promote understanding of threats to home charging infrastructure.
9	Public-private collaboration in the development and operation of CCMTs	Collaboration between clean energy technology developers and government/regulators to address security, crime or safety issues early on. Intelligence-sharing networks between public, private and third sector organisations.
10	Decentralising energy system	Microgrid islanding allows energy systems to operate independently during emergencies or grid disruptions, ensuring a continuous power supply to critical areas.
11	International sanctions and bans	Trade import/ban of CCMTs from regions with bad labour practices. "Decoupling" energy supply from rogue or hostile countries.
12	International cooperation, commitment and assurances	Partnership among the E−7 economies and international partners to share best practices, knowledge, and the transmission of technology. International pledges by nuclear-armed countries (e.g. "no-first-use pledge"), positive security assurances by nuclear-armed states to help those without nuclear weapons if they are under nuclear attack.
13	International law and bodies	International regulation of voluntary carbon credits, international/regional treaty on climate technology. Include crime reporting within Environmental, Social and Governance (ESG) requirements by international frameworks (e.g. Task Force on Climate-Related Financial Disclosures (TCFD)).
14	Regional, national or international policies/laws to define and protect critical infrastructure	Clear, long-term regulatory standards for the definition and protection of national critical infrastructure. Inter-governmental strategy for managing physical infrastructure including monitoring, reporting, and responding; coordinated and unified national response to (cyber and

		information security) threats and incidents against national critical infrastructure, e.g. Computer Security Incident Response Team (CSIRT).
15	Policies and regulation to improve institutional quality, good governance and the rule of law	Anti-corruption frameworks, external auditing and monitoring of CCMT projects by private companies to prevent corrupt practices.
16	International legal structures and frameworks to define and prevent tax avoidance	For example, OECD G20 Base Erosion and Profit Shifting Project (BEPS).
17	Policies and laws to protect human rights of labour force	Policies/legislation against child or illegal labour, and policies/legislation against sexual harassment of female employees.
18	Cybersecurity measures	Anti-spoofing (methods to validate locations of electric vehicles to counter location spoofing), data validation, anomaly detection systems, encryption, false data injection detection, firewall, machine learning, federated learning for privacy-preserving model development, vulnerability simulations and biometric access control. Stronger authentication systems for electric vehicle users charging through smart grid systems: mutual authentication between vehicle-to-grid entities, token-based authentication, anti-hacking protocols/firewalls on networked wind turbines.
19	Radio frequency directed energy weapons (RFDEW)	Using radio frequency weapons to intercept and disrupt criminal or malicious drone activity near CCMT installations.
20	Developing CCMTs with less valuable materials	Use of alternative (i.e., less valuable or less expensive) materials to develop CCMTs to reduce crime risks concerning such materials (e.g. theft, illegal trade and smuggling).
21	Voluntary or mandatory security-by-design approaches	International and domestic legal frameworks that set out 'security-by-design' principles/rules/obligations/standards for industry firms designing/manufacturing CCMTs.
22	Fair compensation for individuals/communities affected by CCMT projects	Fair compensation for individual/communities affected by CCMT installations which rely on their resources (e.g. water or land) to reduce their motivation to commit crimes against the CCMTs or their developers, manufactures and installers.
23	Tag-and-tracing of CCMTs, natural resources and financial/legal instruments	Smart water tagging of critical CCMT (components). Using blockchain technology to tag and trace critical minerals, components and financial/legal instruments (e.g. carbon credits).

24	Obfuscation of open-source mapping of CCMT installations	Obfuscation of Google Maps or other open source geospatial data to prevent criminals from being able to carry out research on their intended targets, particularly on entry/exit routes away from the main road network (e.g. across neighbouring fields).
25	Central information dashboard for government about the decentralised grid	Dashboard to provide governments with a clear view on all assets and real-time information on e.g. generation, distribution and surges.
26	Education and training programs for users and developers of CCMTs	Training and education for manufacturers, sellers, installers and users of CCMTs about their specific crime/security risks, and ways to mitigate them.
27	Semi-formal surveillance	Citizen watchdogs.
28	Incorporating indigenous knowledge in managing or securing CCMTs and related resources	Design out the harmful impact of arson from newly created woodlands by using indigenous knowledge and technology or spatial planning to create breaks in the fire path.
29	Crime risk assessment and crime-proof design of policies and laws	Institutional mechanisms (and related bodies/teams/offices) to assess the crime risks enabled by any relevant policy or regulation (e.g. subsidies for CCMTs or tax discounts etc) and design such policies/regulation so as to minimise such risks.
30	Improved criminal legislation	Define CCMT-specific crimes and improve legislation to address crime in the sector, including on illegal mining, illegal waste disposal/trafficking, fraud and tax evasion.
31	Allocation of part of the budget for government subsidy schemes to fraud detection/prevention	Policies establishing subsidy schemes and other benefits should (pre-emptively) allocate part of the budget to the prevention and detection of fraud.
32	Accreditation and quality/security certification schemes	Quality certifications for installers of CCMTs. Establishing a single point of accreditation for companies that develop, manufacture, install or use CCMTs that can easily be interrogated through an official government website.
33	Specialised independent regulators/auditors and regulatory coordination	Establish green energy regulators. Regulatory bodies to support the integration of high-quality targeted frameworks for extraction industry.
34	Critical mineral waste recovery policies	Improve or introduce domestic policies to maximise recovery of critical minerals that are already in consumer supply chains in the form of “waste”.

35	Increased law enforcement and specialisation	A specialised unit operating across government departments in collaboration with NCA and the police, tasked with anticipating and addressing threats in the CCMT space. Enhanced law enforcement response to illegal labour, illegal mining and trafficking by organised criminal groups through training and capacity building (including through satellite monitoring), stricter policing of ports (for stolen assets that are shipped out of the country), scrap metal dealers (where copper is disposed of) and online market places, and strengthen the role of supranational authorities such as Europol or the EPPO to investigate related crimes independently of national interests.
36	Legal frameworks to allow evidence collection from unmanned drones	Adapted legal framework to allow for unmanned drones (triggered by other local sensing technologies) to collect images of break-ins, theft and burglary that can be used as evidence by law enforcement and courts.